

Lecture 1, part 3 of 3

Model selection, routing, and gateways

Which model, at what cost, for which purpose?

1.2 asked whether you need a model at all. 1.3 assumes the answer came back yes.

Reading a leaderboard without believing it
Benchmarks tell you something. They rarely tell you what
you need.¹

1. Chip Huyen, *AI Engineering* (O'Reilly, 2025), ch. 4: "Model Selection" (p. 179) and "Navigate Public Benchmarks" (p. 191). Longer treatment: Fourrier et al., *The LLM Evaluation Guidebook*. <https://huggingface.co/spaces/OpenEvals/evaluation-guidebook> ↩

Build vs buy

An API call has a price per token. A model you host has a price you pay whether or not anyone calls it.

"Start with inference APIs, but don't be afraid of self-hosting."¹

1. *What We've Learned From A Year of Building with LLMs*, §3.1.3. <https://applied-llms.org/> ↩

Number of models used across experimentation and production



Breakdown by company revenue



Source: a16z survey of 100 CIOs across 15 industries

API Costs (and hidden costs)

Model Routing

Small model by default. Escalate to the big one only when the request needs it.

"Choose the smallest model that gets the job done."¹

1. *What We've Learned From A Year of Building with LLMs*, §2.2.4. <https://applied-llms.org/> ↩

Model Gateways (proxies)

One place in front of every model call.

- **Caching**: the same request twice should cost you once.¹
- **Rate limits**: yours, and the provider's.
- **Cost tracking**: per team, per feature, per request.
- **Logging**: every prompt and every response.

1. "Caching is underrated." *What We've Learned From A Year of Building with LLMs*, §1.3.4. <https://applied-llms.org/> ↩

RECAP: LECTURE 1

1. MLE is building ML systems:
problem, impact, deployment.
2. Law of the instrument: earn the
model before you use it.
3. Choose, route, and gateway your
models deliberately.

Next: form your teams and start
hunting for a problem worth
solving.